

STATISTICAL REVIEW AND EVALUATION

NDA/BLA #:	BLA 111111 (R pilot 4 submission)
Applicant:	R Consortium's R Submission Working Group
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Date(s):	May, 2026
Objective of the submission	Evaluate the use of WebAssembly and Docker containers as methods for packaging an R Shiny application into a fully self-contained submission package, enabling reviewers to run it easily without configuring complex environments
Location of datasets and programs	BLA111111 0008 and 0009
Reviewed tables and figures	Demographic Table, Kaplan Meier (KM) plot for time to first dermatologic event (TTDE), Primary Table, Efficacy Table, and Visit Completion Table in the Shiny app

Pilot 4 Application Review

The R Consortium's R Submission Working Group submitted Pilot 4 in two phases: the original submission with WebAssembly only in September 2024, and a resubmission with both WebAssembly and Docker containers in June 2025. The objective of these pilot submissions was to evaluate the use of WebAssembly and Docker containers as methods for packaging an R Shiny application (originally developed in Pilot 2) into a fully self-contained submission package. This approach aims to enable reviewers to run the application in a robust and reproducible environment without needing to manually configure software dependencies.

Pilot 4 was developed using R version 4.4.1 on a Linux platform. FDA reviewed the submission using various versions of R (both lower and higher than version 4.4.1) on a Windows platform to test performance across different environments.

Given the novelty of the technology, understanding the underlying processes within the application was challenging. To address this, the working group scheduled learning sessions to provide technical details and practical explanations of WebAssembly and Docker containers. Training materials are available at [link](#)¹ for WebAssembly and [link](#)² for Docker containers.

To evaluate the submission, the FDA review team completed the following tasks:

- Received the electronic submission package in electronic common technical document (eCTD) approved formats
- Installed and loaded open-source R packages used as dependencies of the included Shiny application
- Reviewed the Analysis Data Reviewer's Guide (ADRG)
- Installed Windows Subsystem for Linux (WSL) and Docker Desktop
- Followed steps in the ADRG to run R Shiny application using WebAssembly and Docker containers
- Evaluated whether the procedures for running the application required manual configuration of software dependencies
- Identified, investigated, and documented known issues and suggested feasible solutions when possible

The Pilot 4 submission materials and meeting minutes are publicly available on the [R Submission Working Group website](#)³.

Reasons for Requesting Resubmission

The following issues were identified in the initial submission:

1. File format and packaging issues: The working group compressed all items within a 'programs' folder into a zip file archive, which is not acceptable for general FDA regulatory submissions. The zip file contains the source application files and necessary configuration scripts for creating the WebAssembly application. However, it also includes non-permissible file formats, such as .js (JavaScript files). Refer to the [FDA Specification for File Format Types Using eCTD Specifications](#)⁴ for guidance on acceptable formats. A potential alternative approach is to use the pkglite R package to convert the zip file to a text file format that meets submission requirements.

[Initial submission]

BLA111111 > 0008 > m5 > datasets > rconsortiumpilot4 > analysis > adam > programs >				
Sort View ...				
☐	Name	Date modified	Type	Size
	r4app	9/20/2024 3:11 PM	Compressed (zipp...	152,957 KB

¹ https://rpodcast.quarto.pub/submissions-wg-pilot-4/webr_session.html

² https://rpodcast.quarto.pub/submissions-wg-pilot-4/tech_session.html

³ <https://rconsortium.github.io/submissions-wg/>

⁴ [FDA Specification for File Format Types Using eCTD Specifications](#)

[Directory Structure of r4app.zip]

<pre> r4app.zip ├── app │ ├── app.R │ ├── logic │ ├── views │ └── www ├── app_bundle │ ├── README.md │ └── shinyapp.zip ├── renv │ ├── activate.R │ ├── settings.json │ ├── .Rprofile │ ├── renv.lock │ ├── submissions-pilot4-webr.Rproj │ └── utils.R </pre>	r4app > Search r4app <table> <thead> <tr> <th>Name</th><th>Type</th></tr> </thead> <tbody> <tr> <td>app</td><td>File folder</td></tr> <tr> <td>app_bundle</td><td>File folder</td></tr> <tr> <td>renv</td><td>File folder</td></tr> <tr> <td>.Rprofile</td><td>RPROFILE File</td></tr> <tr> <td>renv.lock</td><td>LOCK File</td></tr> <tr> <td>submissions-pilot4-webr</td><td>R Project</td></tr> <tr> <td>utils</td><td>R File</td></tr> </tbody> </table>	Name	Type	app	File folder	app_bundle	File folder	renv	File folder	.Rprofile	RPROFILE File	renv.lock	LOCK File	submissions-pilot4-webr	R Project	utils	R File
Name	Type																
app	File folder																
app_bundle	File folder																
renv	File folder																
.Rprofile	RPROFILE File																
renv.lock	LOCK File																
submissions-pilot4-webr	R Project																
utils	R File																

[Resubmission of Docker container with alternative method using pkglite]

BLA111111 > 0009 > m5 > datasets > rconsortiumpilot4container > analysis > adam > programs

...

☐ Name	Date modified	Type	Size
 dockerfile	6/13/2025 1:15 PM	Text Document	2 KB
 r1pkg	6/13/2025 1:15 PM	Text Document	2,316 KB

[Resubmission of WebAssembly with alternative method using pkglite]

BLA111111 > 0009 > m5 > datasets > rconsortiumpilot4webr > analysis > adam > programs

▼

...

☐	Name	Date modified	Type	Size
	pilot4-webr-pkglite	6/13/2025 1:15 PM	Text Document	13,263 KB
	pkglite-utils	6/13/2025 1:15 PM	R File	6 KB

- Typos and mismatching directory structures and file names compared to the [initial version of Pilot 4 Submission GitHub repository](#)⁵: Typos and discrepancies in directory structures were identified and resolved in the resubmission. Additionally, throughout the Pilot projects, we have noticed discrepancies in file names between what the working group submitted and what FDA reviewers observed in the received submissions. While this is a minor issue, it requires unnecessary modifications when running R code copied directly from the ADRG. Based on our investigation, the following summary addresses file naming considerations for R-based submissions:

⁵ <https://github.com/RConsortium/submissions-pilot4-webr-to-fda/tree/d383de6f29516eabebb395d88e0faf499347d86f>

- FDA Guidance Requirements: Based on the [FDA eCTD guidance](#)⁶, folder and file names must use only letters, numbers, hyphens, or underscores. Blank spaces or special characters, such as periods, are not permitted.
- Pilot 3 Findings: The use of zip files for delivering R packages has been recently approved. R package zip files typically follow a naming convention that combines the package name and version number (major, minor, and patch) separated by periods and an underscore (e.g., packagename_1.0.0.zip). However, because FDA does not allow periods to be used in the folder and file names, R package zip files must be renamed accordingly to comply with submission requirements.
- Pilot 4 Findings: The working group listed underscores in document titles, while the actual physical files contained hyphens. The FDA eCTD team confirmed that no alterations were made by FDA during the submission process. The discrepancies occurred due to the eCTD publishing tool used by the group. The Index.xml is created by the eCTD publishing tool before submission to FDA. Depending on the company's eCTD publishing tool and version, discrepancies in file and folder names may occur between what the applicant submits and what FDA reviewers see. Pilot 4 was submitted using an old eCTD publishing tool that automatically converts underscores to hyphens during submission. To minimize such discrepancies for reviewers, we recommend using an appropriate eCTD publishing tool.

Programs file names in GitHub	 pilot4_webR_pkglite.txt  pkglite_utils.r
Programs file names in eCTD Submission	 pilot4-webr-pkglite  pkglite-utils

BLA111111 > 0009 >				
...				
☐ Name	Date modified	Type	Size	
 m1	6/13/2025 5:02 PM	File folder		
 m5	6/16/2025 8:59 AM	File folder		
 util	6/13/2025 5:03 PM	File folder		
 index	6/13/2025 1:15 PM	Microsoft Edge H...	11 KB	
 index-md5	6/13/2025 1:15 PM	Text Document	1 KB	

⁶ <https://www.fda.gov/media/135373/download>

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE ectd SYSTEM "util/dtd/ich-ectd-3-2.dtd">
<?xml-stylesheet href="util/style/ectd-2-0.xsl" type="text/xsl"?>
<ectd:ectd dtd-version="3.2" xmlns:ectd="http://www.ich.org/ectd" xmlns:xlink="http://www.w3c.org/1999/xlink">
  <m1-administrative-information-and-prescribing-information>
    <leaf ID="ae3068e6df587e5883bdf4462d582ecdd"
      checksum="0eae42f049f26a7f5c6ffb6ed5814058"
      checksum-type="md5" operation="new"
      xlink:href="m1/us/us-regional.xml" xlink:type="simple" xmlns:xlink="http://www.w3c.org/1999/xlink">
      <title>Administrative Information</title>
    </leaf>
  </m1-administrative-information-and-prescribing-information>
  <m5-clinical-study-reports>
    <m5-3-clinical-study-reports>
      <m5-3-5-reports-of-efficacy-and-safety-studies indication="rconsortiumipilot">
        <m5-3-5-4-other-study-reports>
          <leaf ID="a868eaaae7cb63b59429eea0dd1fc2286"
            checksum="58c484cf1571a9f528bd7ea5ab464db4"
            checksum-type="md5" operation="new"
            version="STF Version 2.2"
            xlink:href="m5/53-clin-stud-rep/535-rep-effic-safety-stud/rconsortiumipilot/5354-other-stud-rep/rconsortiumipilot4-
            xlink:type="simple" xmlns:xlink="http://www.w3c.org/1999/xlink">
            <title>rconsortiumipilot4-webr-Jun2025</title>
          </leaf>
          <leaf ID="a5d342e4e8358268e0cd23fb841657078"
            checksum="ae0545102a97321bb7aedb937149b80f"
            checksum-type="md5" operation="new"
            xlink:href="m5/datasets/rconsortiumipilot4-webr/analysis/adam/programs/pkglite-utils.r"
            xlink:type="simple" xmlns:xlink="http://www.w3c.org/1999/xlink">
            <title>pkglite_utils.r</title>
          </leaf>
          <leaf ID="a9d9b767d6b081c75cc751ff12f3c620c"
            checksum="955c9a4c5794317588953665c37eb2ed"
            checksum-type="md5" operation="new"
            xlink:href="m5/datasets/rconsortiumipilot4-webr/analysis/adam/programs/pilot4-webr-pkglite.txt"
            xlink:type="simple" xmlns:xlink="http://www.w3c.org/1999/xlink">
            <title>pilot4_webR_pkglite.txt</title>
          </leaf>
        </m5-3-5-4-other-study-reports>
      </m5-3-5-reports-of-efficacy-and-safety-studies>
    </m5-3-clinical-study-reports>
  </m5-clinical-study-reports>
</ectd:ectd>
```

3. Incorrect hyperlinks: Hyperlinks for software installation were incorrect in the original submission but were updated in the resubmission. Given that the software versions (including R and other dependencies) appear critical for reproducing the same environment, providing accurate hyperlinks and clear specifications is essential.
4. Incomplete directory path in ADRG: The directory path was not fully displayed in the ADRG, which could prevent users from properly following the ADRG instructions. The issues were resolved in the resubmission.

```
# unpack submission pkglite bundle
unpack_pkglite_bundle(
  pkglite_file = "C:/pilot4webR/m5/datasets/rconsortiumipilot4webR/analysis/adam/programs/pilot4_webR_p
  datasets_dir = "C:/pilot4webR/m5/datasets/rconsortiumipilot4webR/analysis/adam/datasets",
  output_dir = "C:/pilot4_webR_files"
)
```

Summary of Testing WebAssembly

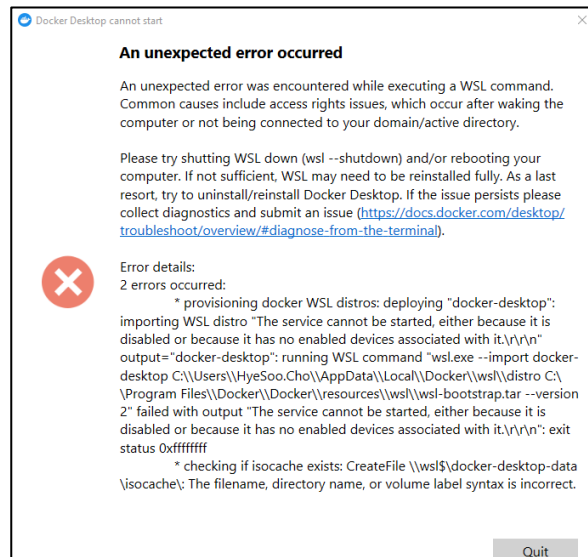
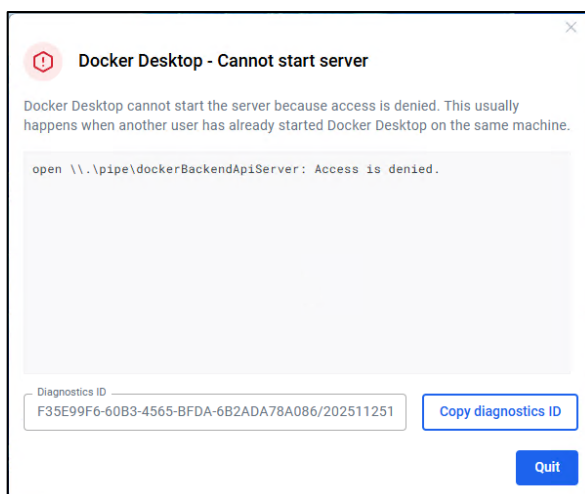
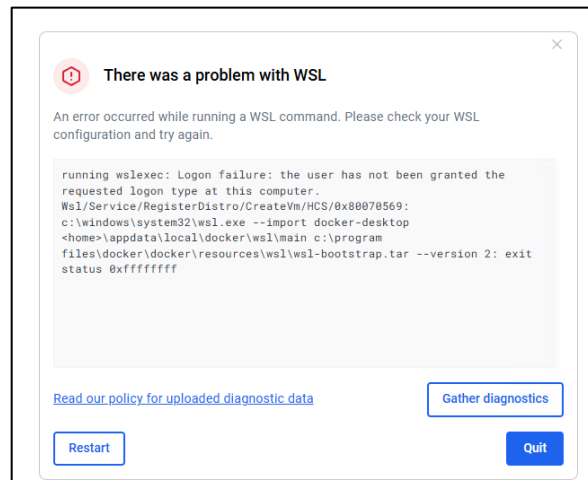
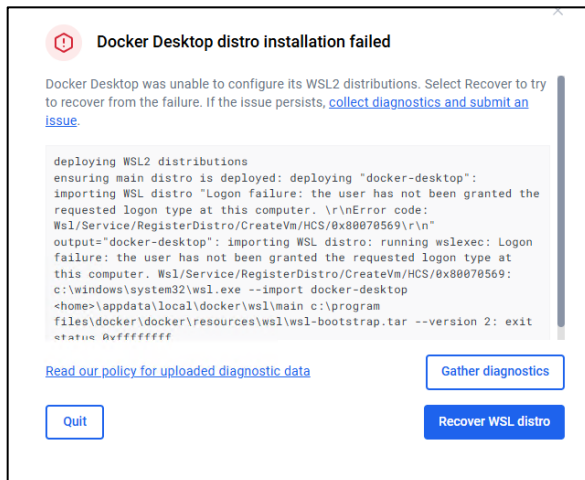
1. No major issues have been identified.
2. Underscore characters ('_') in the original file names were automatically converted to hyphens ('-'). The discrepancies occurred due to the eCTD publishing tool used by the group. Refer to the 'Reasons for Requesting Resubmission' section.
3. R uses forward slashes (/) for directory paths, which differs from the Windows convention of backslashes (\). When you copy a file path from Windows File Explorer, it will automatically include

backslashes (\). Converting these backslashes to forward slashes or escaping them by doubling them in R code is necessary to prevent errors.

Summary of Testing Docker Containers

Docker provides a software engine to construct containers, packaged in the app Docker Desktop. Containers are functional, portable, and self-contained computing environments that rely on the hardware and operating system of the hosting computer.

1. The usability of Docker containers in the FDA environment remains uncertain for the following two reasons:
 - Docker Desktop requires WSL, which may not be permitted on some FDA computing environments, depending on security policies.
 - FDA does not support the installation of Docker Desktop on agency-issued Windows laptops without a paid license or Docker CE (Community Edition).
2. The performance of the software is inconsistent. The following issues occurred when building the containers:



3. The container build fails if Docker Desktop is not running. Each computer restart stops Docker Desktop, so users must restart Docker Desktop after each restart. This prerequisite may not be obvious to users and should be clearly communicated.

```
PS C:\Users\Gabriel.Krotkov> Set-Location -Path "C:\pilot4container_files"
PS C:\pilot4container_files> docker build `
>> --build-arg LOCAL_DATA_DIR=datasets `
>> --build-arg LOCAL_APP_DIR=pilot2wrappers `
>> -t RConsortium/submissions-pilot4-container:latest `
>> -f dockerfile.txt .
ERROR: error during connect: Head "http://%2F%2F.%2Fpipe%2FdockerDesktopLinuxEngine/_ping": open //./pipe/dockerDesktopLinuxEngine: The system cannot find the file specified.
```

4. The standard Docker naming convention is 'Dockerfile' (with a capital 'D'), which does not require a period ('.'). However, FDA submission requirements mandate that file names use lowercase letters. As a result, the file was submitted as 'dockerfile,' which deviates from the standard convention. When using this non-standard name, it's important to include a period in the Docker build command to ensure proper execution. A common error occurs when users omit the final period in the command when working with filenames other than the default 'Dockerfile.'

```
PS C:\pilot4container_files> docker build `
>> --build-arg LOCAL_DATA_DIR=datasets `
>> --build-arg LOCAL_APP_DIR=pilot2wrappers `
>> -t RConsortium/submissions-pilot4-container:latest `
>> -f Dockerfile.txt .
[+] Building 0.1s (1/1) FINISHED                                docker:desktop-linux
=> [internal] load build definition from Dockerfile.txt         0.0s
=> => transferring dockerfile: 2B                               0.0s
ERROR: failed to build: failed to solve: failed to read dockerfile: open Dockerfile.txt: no such file or directory
PS C:\pilot4container_files> docker build `
>> --build-arg LOCAL_DATA_DIR=datasets `
>> --build-arg LOCAL_APP_DIR=pilot2wrappers `
>> -t RConsortium/submissions-pilot4-container:latest `
>> -f dockerfile.txt .
[+] Building 1.1s (12/12) FINISHED                                docker:desktop-linux
=> [internal] load build definition from dockerfile.txt         0.0s
=> => transferring dockerfile: 1.62kB                           0.0s
=> [internal] load metadata for docker.io/rocker/r-ver:4.2.0   0.6s
=> [internal] load .dockerignore                                0.0s
=> => transferring context: 2B                                    0.0s
=> [1/7] FROM docker.io/rocker/r-ver:4.2.0@sha256:53ae350adb0d7807533bc27026923ef1a02bcd9b15e7df016a48b9ca9a003b 0.0s
=> => resolve docker.io/rocker/r-ver:4.2.0@sha256:53ae350adb0d7807533bc27026923ef1a02bcd9b15e7df016a48b9ca9a003b 0.0s
=> [internal] load build context                                0.0s
```

```
PS C:\pilot4container_files> docker build `
>> --build-arg LOCAL_DATA_DIR=datasets `
>> --build-arg LOCAL_APP_DIR=pilot2wrappers `
>> -t RConsortium/submissions-pilot4-container:latest `
>> -f dockerfile.txt
ERROR: docker: 'docker buildx build' requires 1 argument

Usage:  docker buildx build [OPTIONS] PATH | URL | -

Run 'docker buildx build --help' for more information
PS C:\pilot4container_files> docker build `
>> --build-arg LOCAL_DATA_DIR=datasets `
>> --build-arg LOCAL_APP_DIR=pilot2wrappers `
>> -t RConsortium/submissions-pilot4-container:latest `
>> -f dockerfile.txt .
[+] Building 1.0s (12/12) FINISHED                                docker:desktop-linux
=> [internal] load build definition from dockerfile.txt         0.1s
=> => transferring dockerfile: 1.62kB                           0.0s
=> [internal] load metadata for docker.io/rocker/r-ver:4.2.0   0.5s
=> [internal] load .dockerignore                                0.0s
=> => transferring context: 2B                                    0.0s
=> [internal] load build context                                0.1s
=> => transferring context: 2.90kB                               0.0s
=> [1/7] FROM docker.io/rocker/r-ver:4.2.0@sha256:53ae350adb0d7807533bc27026923ef1a02bcd9b15e7df016a48b9ca9a003b 0.0s
=> => resolve docker.io/rocker/r-ver:4.2.0@sha256:53ae350adb0d7807533bc27026923ef1a02bcd9b15e7df016a48b9ca9a003b 0.0s
=> CACHED [2/7] RUN apt-get update --quiet && apt-get install curl libssl-dev libcurl4-openssl 0.0s
=> CACHED [3/7] RUN useradd -m shiny                             0.0s
=> CACHED [4/7] COPY pilot2wrappers /home/shiny/submissions-pilot2 0.0s
=> CACHED [5/7] COPY datasets /home/shiny/submissions-pilot2/datasets 0.0s
=> CACHED [6/7] WORKDIR /home/shiny/submissions-pilot2          0.0s
=> CACHED [7/7] RUN Rscript -e "options('renv.config.install.remotes' = FALSE)" -e "renv::restore()" 0.0s
```

5. A Docker container repository name containing uppercase letters resulted in intermittent errors during the review process. A potential recommendation is to require all lowercase letters for Docker repository names.

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\HyeSoo.Cho> Set-Location -Path "C:/Users/HyeSoo.Cho/Downloads/pilot4container_files"
PS C:\Users\HyeSoo.Cho\Downloads\pilot4container_files> docker build `
>> --build-arg LOCAL_DATA_DIR=datasets `
>> --build-arg LOCAL_APP_DIR=pilot2wrappers `
>> -t RConsortium/submissions-pilot4-container:latest `
>> -f Dockerfile.txt .
invalid argument "RConsortium/submissions-pilot4-container:latest" for "-t, --tag" flag: invalid reference format: repository
name must be lowercase
See 'docker build --help'.
PS C:\Users\HyeSoo.Cho\Downloads\pilot4container_files> docker build `
>> --build-arg LOCAL_DATA_DIR=datasets `
>> --build-arg LOCAL_APP_DIR=pilot2wrappers `
>> -t submissions-pilot4-container:latest `
>> -f Dockerfile.txt .
[+] Building 1.6s (12/12) FINISHED
=> [internal] load build definition from Dockerfile.txt                                0.0s
=> => transferring dockerfile: 1.62kB                                                0.0s
=> [internal] load .dockerignore                                                       0.0s
=> => transferring context: 2B                                                        0.0s
=> [internal] load metadata for docker.io/rocker/r-ver:4.2.0                        1.3s
=> [1/7] FROM docker.io/rocker/r-ver:4.2.0@sha256:53ae350adb0d7807533bc27026923ef1a02bcd9b15e7df016a48b9ca9a003b 0.0s
=> [internal] load build context                                                       0.1s
=> => transferring context: 20.14MB                                                  0.1s
=> CACHED [2/7] RUN apt-get update --quiet && apt-get install curl libssl-dev libcurl4-openssl 0.0s
=> CACHED [3/7] RUN useradd -m shiny                                                  0.0s
=> CACHED [4/7] COPY pilot2wrappers /home/shiny/submissions-pilot2                  0.0s
=> CACHED [5/7] COPY datasets /home/shiny/submissions-pilot2/datasets              0.0s
=> CACHED [6/7] WORKDIR /home/shiny/submissions-pilot2                             0.0s
=> CACHED [7/7] RUN Rscript -e "options('renv.config.install.remotes' = FALSE)" -e "renv::restore()" 0.0s
=> => exporting to image                                                            0.0s
=> => exporting layers                                                            0.0s
=> => writing image sha256:e88c636dbc204a1ce26e5de40aa49c72b4cd2c80505699be30fb4101a97a5f1d 0.0s
=> => naming to docker.io/library/submissions-pilot4-container:latest              0.0s
PS C:\Users\HyeSoo.Cho\Downloads\pilot4container_files> docker run -it --rm -p 8787:8787 submissions-pilot4-container:latest
```

6. The terminal displayed `http://0.0.0.0:8787` as the URL to access the Shiny application running in the Docker container; however, the application could only be accessed using `http://127.0.0.1:8787` instead. To access the Shiny application running in the Docker container, two key R options were configured: `shiny.port = 8787` to designate port 8787, and `shiny.host = '0.0.0.0'` to make the application accessible from the host machine's network (since Docker containers operate on an isolated network that is otherwise inaccessible from the host system). On Linux systems, the application can be accessed via `http://0.0.0.0:8787`, which automatically resolves to `http://127.0.0.1:8787`; however, on Windows systems running Docker through WSL, users must enter `http://127.0.0.1:8787` directly due to an additional network abstraction layer. Finally, the Docker run command option `-p 8787:8787` maps the container's port 8787 to the host machine's port 8787, without which the application would not be accessible outside the container.

```
* Project '~/submissions-pilot2' loaded. [renv 0.15.2]
> options(shiny.port = 8787, shiny.host = '0.0.0.0'); pkgload::load_all(export_all = FALSE, helpers = FALSE, attach_testthat =
FALSE);options('golem.app.prod' = TRUE); pilot2wrappers::run_app()
[1] Loading pilot2wrappers
Loading required package: shiny
Listening on http://0.0.0.0:8787
```

7. R versions inconsistent with the ADRG specification may encounter errors when installing necessary R packages during the restoration process using renv R package.

```
R version 4.3.1 (2023-06-16 ucrt) -- "Beagle Scouts"
Copyright (C) 2023 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64 (64-bit)

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

i using R 4.3.1 (lockfile was generated with R 4.4.3)
- Project 'C:/Users/HyeSoo.Cho/Downloads/pilot4webR_files' loaded. [renv 1.0.7]
- One or more packages recorded in the lockfile are not installed.
- Use 'renv::status()' for more details.
> renv::restore(prompt = FALSE)
The following package(s) will be updated:

# Bioconductor 3.19 -----
- BioVersion      [* -> 3.19.1]

# CRAN -----
- BiocManager     [1.30.27 -> 1.30.23]
- lattice         [0.21-8 -> 0.22-6]
- MASS            [7.3-60 -> 7.3-60.2]
- Matrix          [1.5-4.1 -> 1.7-0]
- mgcv            [1.8-42 -> 1.9-1]
- nlme            [3.1-162 -> 3.1-164]
- archive         [* -> 1.1.8]
- askpass         [* -> 1.2.0]

- tinytex        [* -> 0.51]
- tzd            [* -> 0.4.0]
- utf8           [* -> 1.2.4]
- vctrs          [* -> 0.6.5]
- viridisLite    [* -> 0.4.2]
- visr           [* -> 0.4.1]
- vroom          [* -> 1.6.5]
- withr          [* -> 3.0.0]
- xfun           [* -> 0.45]
- xtable         [* -> 1.8-4]
- yaml           [* -> 2.3.8]

# Downloading packages -----
- Downloading MASS from CRAN ... ERROR [unknown reason]
- Downloading MASS from CRAN ... ERROR [unknown reason]
- Downloading MASS from CRAN ... ERROR [unknown reason]
- Downloading MASS from CRAN ... ERROR [unknown reason]
- Downloading MASS from CRAN ... ERROR [unknown reason]
- Downloading MASS from CRAN ... OK [file is up to date]
- Downloading BiocManager from P3M ... OK [file is up to date]
- Downloading Matrix from CRAN ... ERROR [unknown reason]
- Downloading Matrix from CRAN ... ERROR [unknown reason]
- Downloading Matrix from CRAN ... ERROR [unknown reason]
- Downloading Matrix from CRAN ... ERROR [unknown reason]
- Downloading Matrix from CRAN ... ERROR [unknown reason]
- Downloading Matrix from CRAN ... OK [file is up to date]
- Downloading lattice from P3M ... OK [file is up to date]
- Downloading mgcv from P3M ... OK [file is up to date]
- Downloading nlme from P3M ... OK [file is up to date]
- Downloading survival from P3M ... OK [file is up to date]
- Downloading Biocversion from Bioconductor 3.19 ... ERROR [unknown reason]
- Downloading Biocversion from Bioconductor 3.19 ... ERROR [unknown reason]
- Downloading Biocversion from Bioconductor 3.19 ... ERROR [unknown reason]
- Downloading Biocversion from Bioconductor 3.19 ... ERROR [unknown reason]
- Downloading Biocversion from Bioconductor 3.19 ... ERROR [unknown reason]
warning: failed to find binary for 'Biocversion 3.19.1' in package repositories
warning: failed to find source for 'Biocversion 3.19.1' in package repositories
warning: error downloading 'https://bioconductor.org/packages/3.19/bioc/src/contrib/Archive/Biocversion/Biocversion_3.19.1.tar.gz' [unknown reason]
warning: error downloading 'https://bioconductor.org/packages/3.19/data/annotation/src/contrib/Archive/Biocversion/Biocversion_3.19.1.tar.gz' [unknown reason]
warning: error downloading 'https://bioconductor.org/packages/3.19/data/experiment/src/contrib/Archive/Biocversion/Biocversion_3.19.1.tar.gz' [unknown reason]
warning: error downloading 'https://bioconductor.org/packages/3.19/workflows/src/contrib/Archive/Biocversion/Biocversion_3.19.1.tar.gz' [unknown reason]
warning: error downloading 'https://bioconductor.org/packages/3.19/books/src/contrib/Archive/Biocversion/Biocversion_3.19.1.tar.gz' [unknown reason]
warning: error downloading 'https://p3m.dev/cran/latest/src/contrib/Archive/Biocversion/Biocversion_3.19.1.tar.gz' [unknown reason]
Error: failed to retrieve package 'Biocversion@3.19.1'
> source("utils.R")
> build_app()

Error in loadNamespace(x) : there is no package called 'fs'
```

Conclusion

WebAssembly and Docker containers represent viable approaches for delivering R Shiny applications. However, based on the Pilot 4 review, these methods may not be the most efficient or robust options for FDA regulatory submissions.

WebAssembly: Reproducing the computational environment—particularly maintaining R package version control—remains essential for FDA submissions utilizing R. Even when implementing WebAssembly, the renv package was still required to restore packages required to build the WebAssembly application, as well as the execution of a web server process to view the application. These

requirements indicate that this approach does not eliminate the fundamental challenges of reproducing the same environment as the applicant to ensure the application can be executed successfully.

Docker containers: The installation of WSL and Docker software posed technical challenges during the review process. Furthermore, the compatibility of Docker containers with the FDA computing environment remains unclear and may be constrained by FDA security policies that could restrict practical implementation. Additionally, R version compatibility proved to be a critical factor in successful deployment, indicating that this method does not fully eliminate the need for manual configuration of software dependencies and may not provide the level of robustness required for regulatory submissions.